

AI-Assisted Chest X-ray Interpretation Using Deep Learning, Explainable AI (Grad-CAM), and Retrieval-Augmented Generation (RAG)

Abstract

Chest X-ray imaging is one of the most commonly used diagnostic tools for detecting thoracic diseases. However, manual interpretation of chest X-rays is time-consuming and requires significant expertise. This project presents an AI-assisted chest X-ray interpretation system using Deep Learning, Explainable Artificial Intelligence (XAI), and Retrieval-Augmented Generation (RAG). The NIH ChestX-ray14 dataset was used to train and evaluate disease classification models. A custom Convolutional Neural Network (CNN) and a transfer learning-based ResNet18 model were implemented and compared. Model explainability was achieved using Grad-CAM, which highlights image regions influencing predictions. A lightweight RAG framework was incorporated to provide grounded medical explanations based on retrieved knowledge. Experimental results demonstrated that ResNet18 achieved a Mean AUROC of 0.7340, outperforming the custom CNN model with a Mean AUROC of 0.6741. The proposed system demonstrates the feasibility of combining deep learning, explainability, and retrieval-based reasoning to support chest X-ray interpretation.

Keywords: Chest X-ray, Deep Learning, ResNet18, CNN, Grad-CAM, Explainable AI, Retrieval-Augmented Generation, Medical Imaging.

1. Introduction

Medical imaging plays a critical role in diagnosing various diseases. Chest X-rays are among the most frequently performed radiological examinations worldwide. They are used to detect conditions such as pneumonia, pneumothorax, cardiomegaly, edema, and pleural effusion.

Despite their widespread use, interpreting chest X-rays can be challenging due to subtle visual differences between disease patterns. Recent advancements in Artificial Intelligence and Deep Learning have enabled automated analysis of medical images, assisting healthcare professionals in disease detection.

This project aims to develop an AI-assisted chest X-ray interpretation system capable of:

- Detecting multiple thoracic diseases.
- Providing explainable predictions using Grad-CAM.

- Generating grounded disease explanations using RAG.
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2. Problem Statement

Manual chest X-ray interpretation requires significant clinical expertise and can be subject to inter-observer variability. The objective of this project is to develop a deep learning-based system capable of:

1. Automatically classifying chest X-ray images.
 2. Identifying 14 thoracic disease categories.
 3. Explaining predictions using visual evidence.
 4. Providing grounded textual interpretations through retrieval-based reasoning.
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3. Dataset Description

The NIH ChestX-ray14 dataset was used in this project.

Dataset Characteristics

- Total Images: 112,120
- Unique Patients: 30,805
- Disease Categories: 14
- Image Format: PNG
- Image Type: Chest X-ray

Disease Labels

The dataset contains the following disease classes:

1. Atelectasis
2. Cardiomegaly
3. Effusion
4. Infiltration
5. Mass
6. Nodule
7. Pneumonia
8. Pneumothorax
9. Consolidation
10. Edema
11. Emphysema

- 12. Fibrosis
- 13. Pleural Thickening
- 14. Hernia

Dataset Split

Official NIH-provided train and test lists were used.

- Train + Validation Images: 86,524
- Test Images: 25,596

Figure 1

Disease Distribution Output

Finding Labels		count
No Finding		2754
Infiltration		417
Atelectasis		192
Effusion		156
Nodule		106
Fibrosis		94
Pneumothorax		90
Cardiomegaly		72
Consolidation		64
Atelectasis Infiltration		62
Pleural_Thickening		62
Mass		60
Emphysema		55
Effusion Infiltration		50
Atelectasis Effusion		33
Cardiomegaly Effusion		30
Consolidation Infiltration		29
Infiltration Nodule		24
Atelectasis Effusion Infiltration		23
Atelectasis Consolidation		19

4. Data Preprocessing

Several preprocessing steps were applied before model training.

Image Processing

- Images resized to 224×224 pixels.
- Converted to grayscale.
- Converted to tensors.
- Normalized for stable training.

Label Processing

The disease labels were converted into multi-label binary vectors.

Example:

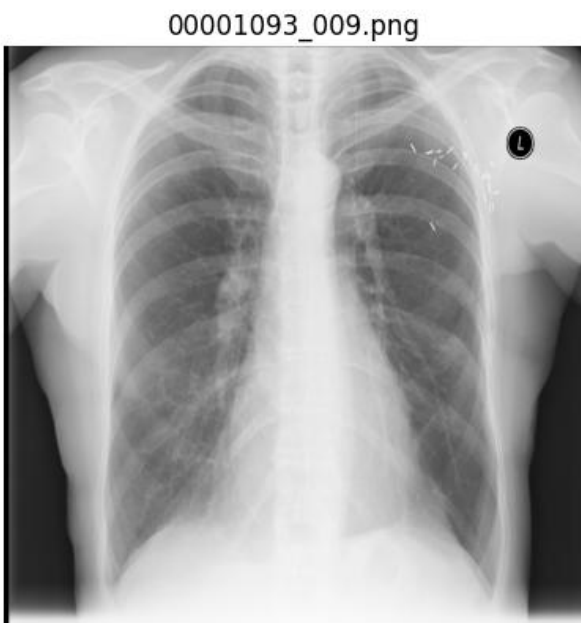
Disease	Binary Encoding
Pneumothorax	1
Others	0

Dataset Sampling

To reduce computational requirements, a subset of 5,000 images was selected for experimentation.

Figure 2

Sample Chest X-ray Image



5. Deep Learning Methodology

5.1 Custom CNN Model

A Convolutional Neural Network (CNN) was developed from scratch.

Architecture

- Convolution Layer 1
- ReLU Activation
- Max Pooling
- Convolution Layer 2
- ReLU Activation
- Max Pooling
- Fully Connected Layer
- Output Layer (14 Classes)

Loss Function

Binary Cross Entropy with Logits Loss (BCEWithLogitsLoss)

Optimizer

Adam Optimizer

Learning Rate

0.001

Training Epochs

5

5.2 ResNet18 Transfer Learning

Transfer Learning was applied using ResNet18 pretrained on ImageNet.

Advantages

- Faster convergence.
- Better feature extraction.
- Improved generalization.

The final classification layer was modified to produce 14 disease outputs.

Learning Rate

0.0001

Training Epochs

5

5.3 VGG19 Transfer Learning

To further evaluate the effectiveness of transfer learning for chest X-ray disease classification, a VGG19 model pretrained on ImageNet was implemented.

Advantages

- Deep architecture capable of learning hierarchical image features.
- Proven performance in medical image classification tasks.
- Effective transfer learning through pretrained convolutional layers.

The final classification layer was modified to output predictions for 14 thoracic disease classes.

Learning Rate

0.0001

Training Epochs

5

Batch Size

16

Loss Function

Binary Cross Entropy with Logits Loss (BCEWithLogitsLoss)

Optimizer

Adam Optimizer

The VGG19 model was trained and evaluated using the same NIH ChestXray14 subset used for the CNN and ResNet18 experiments.

6. Model Training

Custom CNN Training Results

Epoch Training Loss

1	0.1904
2	0.1730
3	0.1708
4	0.1650
5	0.1574

Validation Loss = 0.1849

ResNet18 Training Results

Epoch Training Loss

1	0.2774
2	0.1591
3	0.1373
4	0.1039
5	0.0633

VGG19 Training Results

Epoch Training Loss

1	0.1878
2	0.1746
3	0.1713
4	0.1674

Epoch Training Loss

5 0.1635

The gradual reduction in training loss across all models demonstrates successful learning of disease-related image features.

7. Evaluation Metrics

The following metrics were used.

AUROC

Area Under Receiver Operating Characteristic Curve measures the model's ability to distinguish positive and negative classes.

PR-AUC

Precision-Recall Area Under Curve is particularly useful for imbalanced medical datasets.

Precision

Measures the proportion of correct positive predictions.

Recall

Measures the ability to identify actual positive cases.

F1 Score

Harmonic mean of Precision and Recall.

8. Experimental Results

Custom CNN Performance

Mean AUROC = 0.6741

ResNet18 Performance

Mean AUROC = 0.7340

Mean PR-AUC = 0.1876

VGG19 Performance

Mean AUROC = 0.6674

Mean PR-AUC = 0.1228

Model Comparison

Model	Mean AUROC	Mean PR-AUC
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Custom CNN	0.6741	Not Calculated
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ResNet18	0.7340	0.1876
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VGG19	0.6674	0.1228
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Among all evaluated models, ResNet18 achieved the highest overall performance. The pretrained residual architecture enabled better feature extraction and generalization compared to both the custom CNN and VGG19.

Although VGG19 successfully learned disease representations, its performance remained lower than ResNet18. This may be attributed to the higher parameter count of VGG19 and the limited subset size used during experimentation.

Figure 3

Final Model Comparison Table

	Model	Mean AUROC	Mean PR-AUC
0	Simple CNN	0.6741	Not calculated
1	ResNet18	0.7340	0.1876
2	VGG19	0.6674	0.1228

9. Disease-wise AUROC Analysis

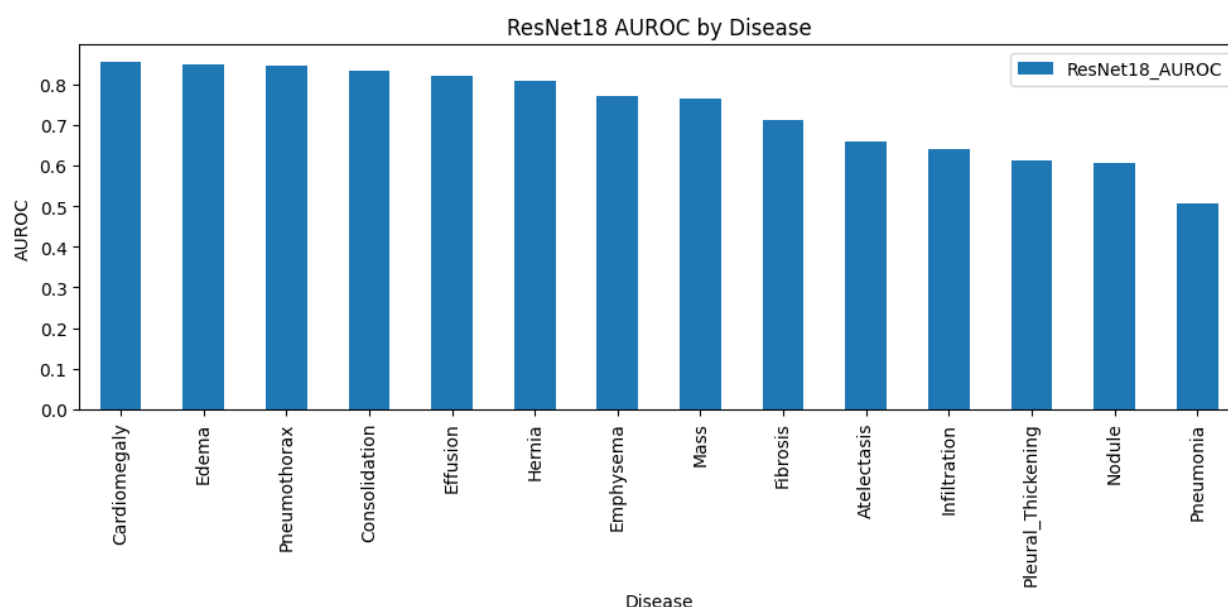
The top-performing diseases were:

Disease	AUROC
Cardiomegaly	0.8544
Edema	0.8482
Pneumothorax	0.8454
Consolidation	0.8322
Effusion	0.8192

The lowest-performing disease was Pneumonia with an AUROC of approximately 0.51.

Figure 4

ResNet18 AUROC by Disease Chart



10. Explainable AI Using Grad-CAM

Deep learning models are often criticized as black-box systems. To improve interpretability, Grad-CAM was employed.

Purpose

Grad-CAM highlights image regions contributing most strongly to the model prediction.

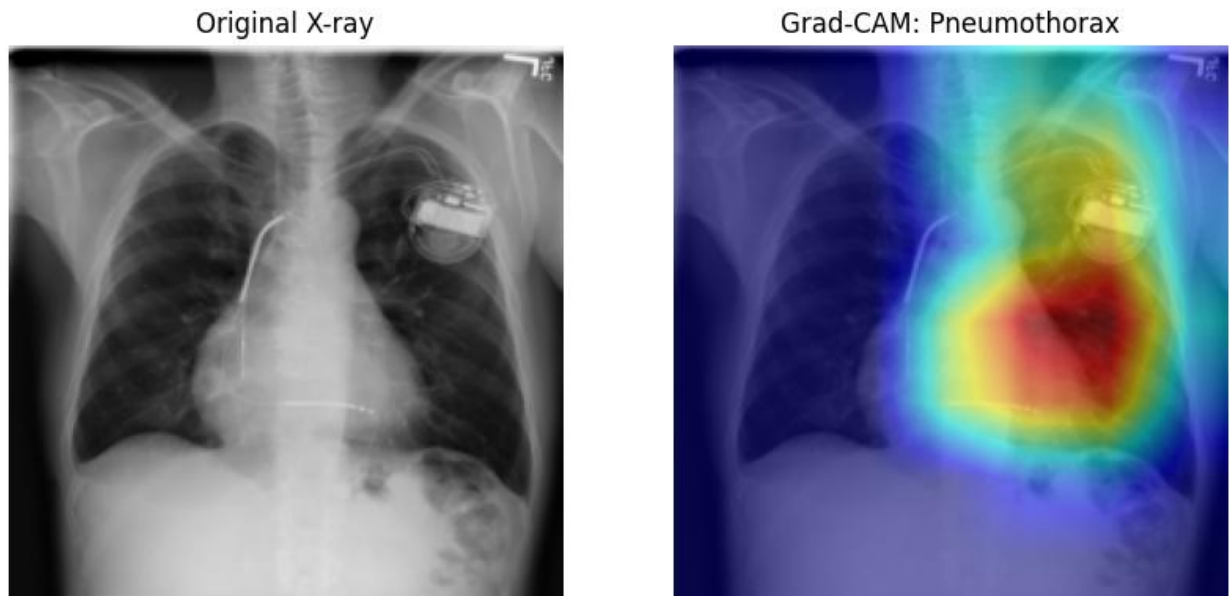
Benefits

- Improves transparency.
- Helps clinicians verify predictions.

- Increases trust in AI systems.

Figure 5

Original X-ray and Grad-CAM Heatmap



The highlighted regions indicate areas that significantly influenced the model's prediction.

11. Retrieval-Augmented Generation (RAG)

A lightweight RAG framework was incorporated to generate grounded explanations.

Workflow

1. Predict disease using ResNet18.
2. Retrieve relevant medical knowledge.
3. Generate disease explanation.
4. Present interpretation with disclaimer.

Example Output

Predicted Disease: Pneumothorax

Explanation: Collapsed lung caused by air leakage.

The retrieved evidence supports the predicted disease and provides additional context.

RAG Interpretation Output

RAG-grounded AI Interpretation

Predicted Finding:

Pneumothorax

Model Confidence:

56.22%

Retrieved Evidence:

[DOC1] Pneumothorax refers to air in the pleural space, which can cause partial or complete lung collapse.

[DOC5] Grad-CAM highlights image regions that contributed most strongly to a CNN model prediction.

[DOC4] The AI model is assistive only and should not be used as a final medical diagnosis.

Interpretation:

The model predicts possible Pneumothorax. Based on retrieved evidence, Pneumothorax refers to air in the pleural space, which can cause partial or complete lung collapse.

The Grad-CAM heatmap provides visual support by highlighting regions that influenced the model prediction.

Disclaimer:

This is an AI-assisted educational output only. It is not a medical diagnosis. A qualified radiologist must make the final interpretation.

12. Conclusion

This project developed an AI-assisted chest X-ray interpretation system that combines Deep Learning, Explainable AI (Grad-CAM), and Retrieval-Augmented Generation (RAG).

Three different deep learning models were evaluated:

- Custom CNN
- ResNet18
- VGG19

Key findings include:

- ResNet18 achieved the highest Mean AUROC of 0.7340.
- VGG19 achieved a Mean AUROC of 0.6674.
- The custom CNN achieved a Mean AUROC of 0.6741.

- ResNet18 also achieved the highest Mean PR-AUC of 0.1876.
- Grad-CAM successfully highlighted image regions influencing model predictions.
- RAG provided grounded medical explanations using retrieved clinical knowledge.
- The system demonstrated the feasibility of combining prediction, explainability, and retrieval-based reasoning in medical imaging workflows.

Overall, ResNet18 was selected as the best-performing model and forms the final backbone of the AI-assisted chest X-ray interpretation system.

The project successfully demonstrates an end-to-end pipeline for automated chest X-ray analysis using modern deep learning and explainable AI techniques.

13. Future Work

Future improvements include:

- Training on the complete NIH dataset.
 - Using advanced architectures such as DenseNet121 and EfficientNet.
 - Integrating Large Language Models (LLMs) for richer explanations.
 - Deploying the system using FastAPI and Streamlit.
 - Integrating experiment tracking using MLflow or Weights & Biases.
 - Real-world deployment in healthcare environments.
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